

# Tools & Utilities

## 1. URL Shortener

### Core Outcomes

- Custom Short Links: Allow custom alias creation
- Link Analytics: Track clicks, referrers, and locations
- Link Management: Dashboard for all shortened URLs
- QR Code Generation: Generate QR codes for shortened URLs
- Expiration Settings: Set link expiration dates

### Technical Skills Gained

- Redirect Handling: Efficient 301/302 redirect implementation
- Analytics Collection: Track and visualize link performance
- Rate Limiting: Prevent abuse of shortening service
- Bulk Operations: Import/export multiple URLs
- API Development: REST API for programmatic access

## 2. File Sharing Service

### Core Outcomes

- Secure Uploads: Password protection and encryption
- Expiration Control: Set automatic file deletion dates
- Large File Support: Chunked uploads for large files
- Preview Generation: File previews without downloading
- Download Analytics: Track file access and downloads

### Technical Skills Gained

- File Processing: Handle various file types and sizes
- Storage Optimization: Implement efficient file storage
- Security: File encryption and access control
- Streaming: Progressive file downloads
- Cleanup Jobs: Automated file deletion scheduling

## 3. Code Snippet Manager

### Core Outcomes

- Syntax Highlighting: Support for all major languages
- Organization: Folders, tags, and search functionality

- Sharing: Public/private snippets with embed codes
- Version History: Track changes to snippets over time
- Collections: Group related snippets into collections

#### **Technical Skills Gained**

- Code Parsing: Language detection and syntax highlighting
- Search Indexing: Fast code search across snippets
- Embed Systems: Create embeddable snippet widgets
- Import/Export: Support for various code formats
- API Development: Programmatic snippet management

## **4. API Testing Dashboard**

#### **Core Outcomes**

- Request Builder: Visual interface for building API requests
- Environment Variables: Manage different API environments
- Test Collections: Organize and run groups of API tests
- Automated Testing: Schedule and automate API tests
- Documentation: Auto-generate API documentation

#### **Technical Skills Gained**

- HTTP Protocol: Deep understanding of HTTP methods and headers
- Authentication: Handle various auth methods (OAuth, API keys, etc.)
- Scripting: Implement pre-request and test scripts
- Monitoring: API uptime and performance monitoring
- Reporting: Generate test execution reports

## **5. Password Manager**

#### **Core Outcomes**

- Secure Storage: Encrypted password database
- Password Generator: Strong password generation
- Auto-fill: Browser extension for password auto-fill
- Security Audit: Identify weak and reused passwords
- Emergency Access: Share access in emergencies

#### **Technical Skills Gained**

- Cryptography: Implement encryption and decryption
- Browser Extensions: Develop cross-browser extensions
- Security Best Practices: Implement security protocols
- Sync Systems: Sync across devices securely
- Biometric Integration: Support for fingerprint/face ID